TEAM ZOLLERQU

Splitting a simulation mesh between a CPU and a GPU

One qubit per mesh region. Run on IonQ's noisy simulator and graded against the exact best split.

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Qollab x IonQ Hackathon, October 2026

This effort is supported by Qollab & IonQ

12 qubits

one per mesh region

1,000 shots

IonQ Forte 1 noise model

27.8 %

are the exact best split

IonQ Forte 1 noise model (Qollab), 11 Oct 2026

THE PROBLEM

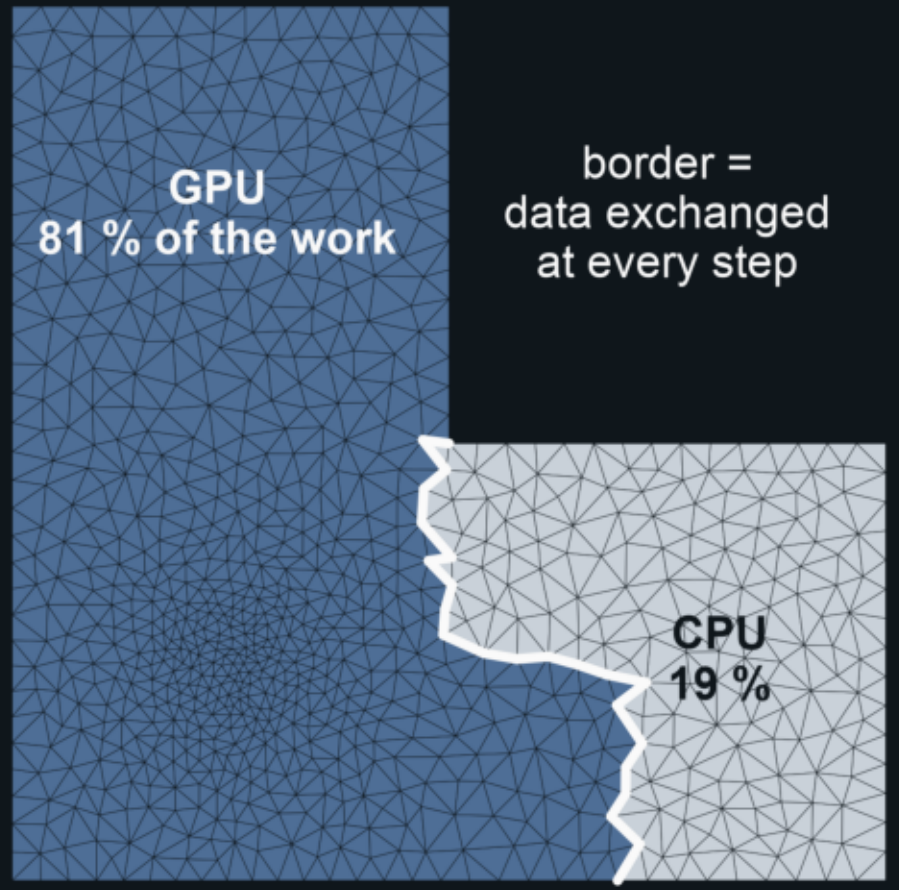
One mesh, two devices: who computes what?

80 % of the work should go to a GPU that is 4x faster than the CPU

assumed device speeds (model)

Who: engineers who run large simulations (heat, stress, crash) on CPU + GPU clusters.

Why: a bad split leaves one device idle, or makes the devices exchange more data.



our mesh (1,670 triangles), the exact best split for CPU + GPU (classical)

FORMULATION

From mesh to Ising energy

$$E(z) = \sum_{\langle i,j \rangle} c_{ij} \frac{1 - z_i z_j}{2} + \lambda \left(\sum_i a_i z_i - (1 - 2\alpha) \right)^2$$

border that is cut

penalty if the work share is off target

1 qubit =
1 mesh region

0: device A 1: device B

12 regions =
4,096 splits

all checked: the best is known

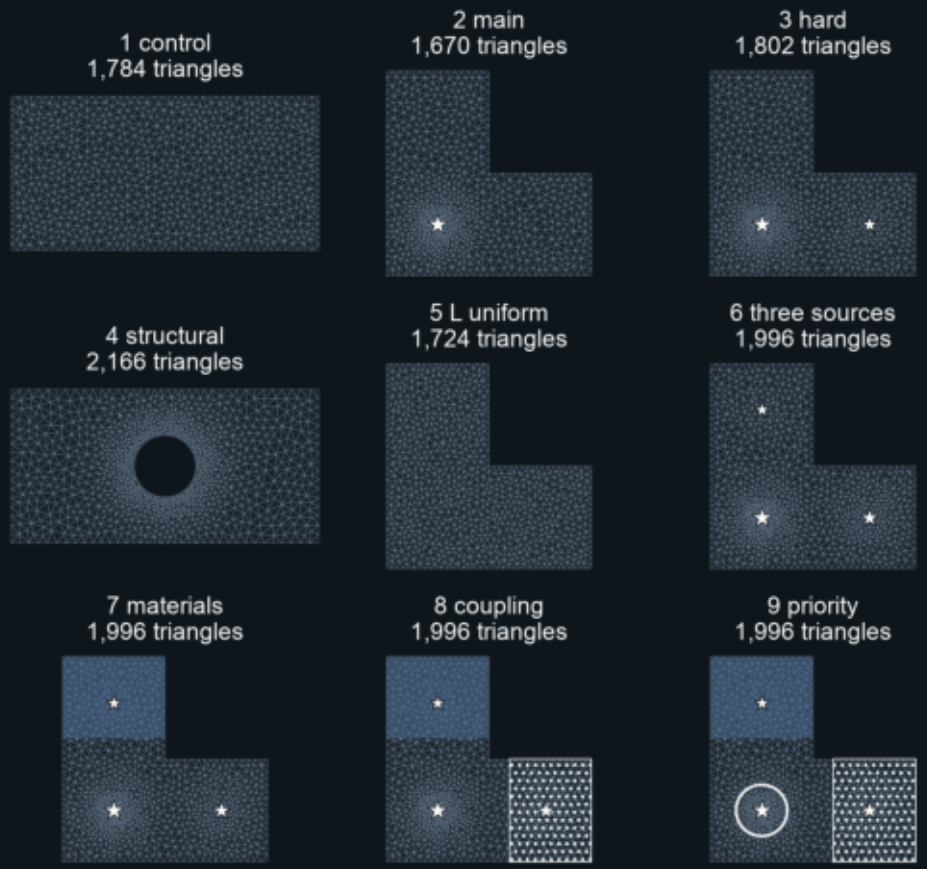
alpha =
target work share

0.5 equal, 0.8 CPU + GPU

exact best split: classical check of all splits

BENCHMARK

9 mesh objects, one method



9 objects, built from 6 different meshes (1,670 to 2,166 triangles)

Shapes: a plate, L-shaped plates and a plate with a hole

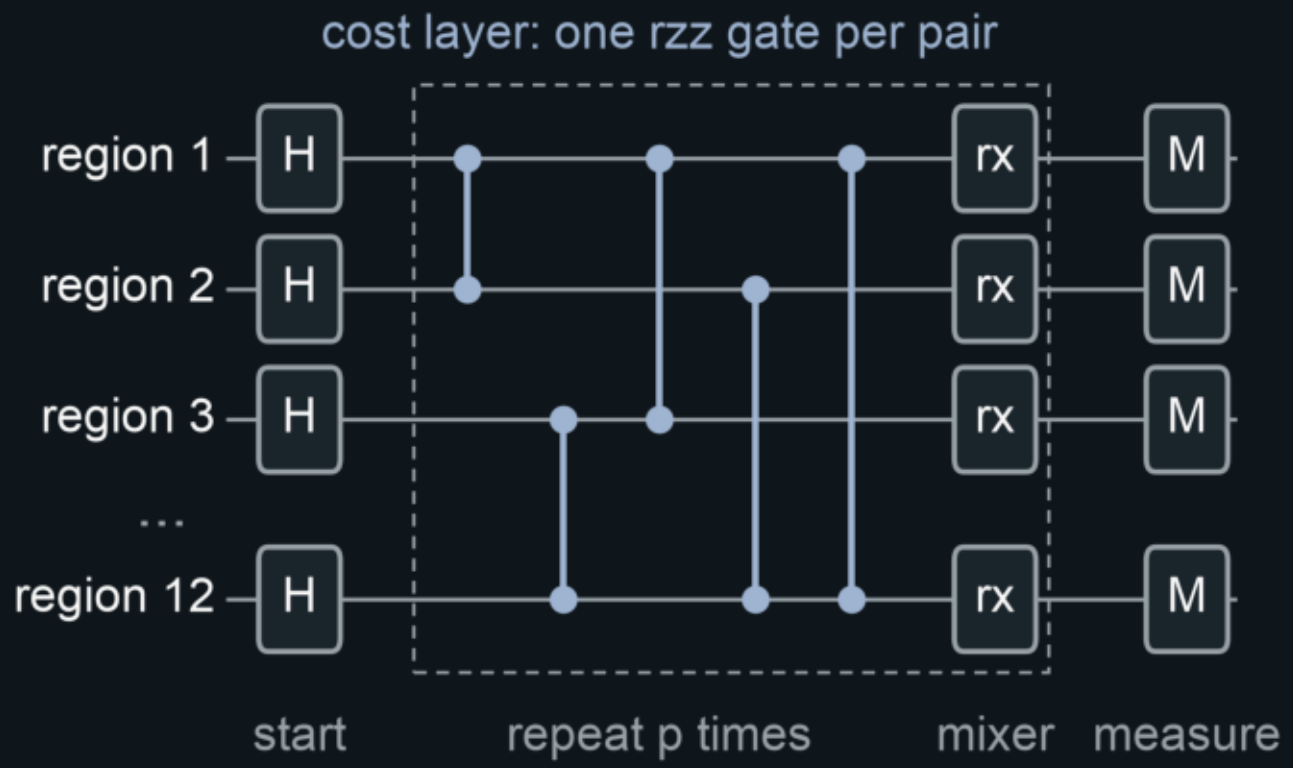
What changes the work: heat sources, heavy material, strongly coupled zones, a priority zone

Every object: both device scenarios, grouped into 12 to 29 regions (one qubit each)

frozen meshes in data/frozen_meshes (identical on every machine)

QUANTUM METHOD

Fixed circuits, trained offline



sketch of the QAOA circuit; varQITE uses the same start and measurement

104 circuits compared at 12 qubits

LR-QAOA, optimised QAOA, CVaR-QAOA, warm starts, varQITE (IonQ's method)

Trained on a noiseless simulator

ONE fixed circuit per IonQ job

noiseless simulation (results/ionq_selection)

WHY IONQ

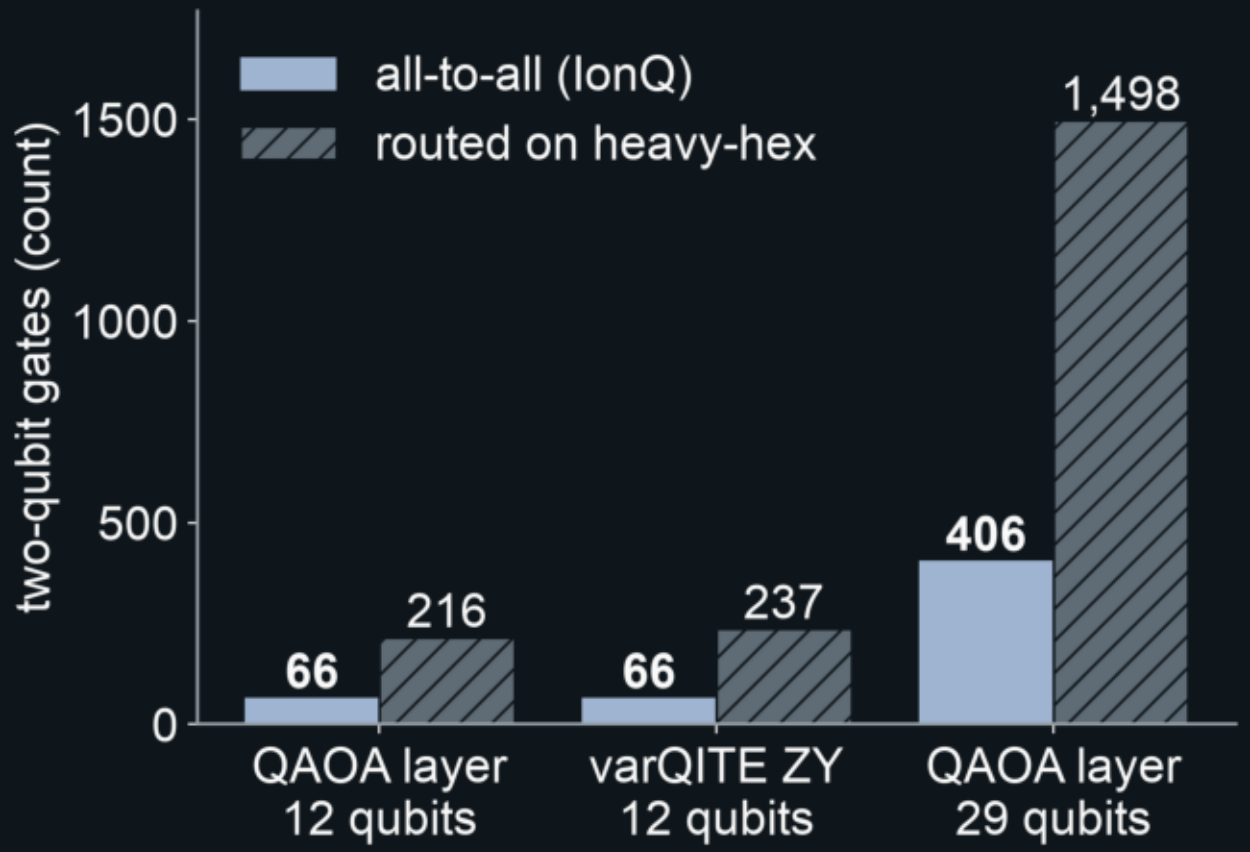
Every region pairs with every other

66 two-qubit gates for one layer at 12 qubits, all-to-all connectivity

216 after routing on a heavy-hex layout: 3.3x more

At 29 qubits: 406 against 1,498 (3.7x more)

counted with the Qiskit transpiler only (no simulation, no hardware run)



MAIN RESULT · 2 EQUAL DEVICES

28 % of shots are the exact best split

27.8 %

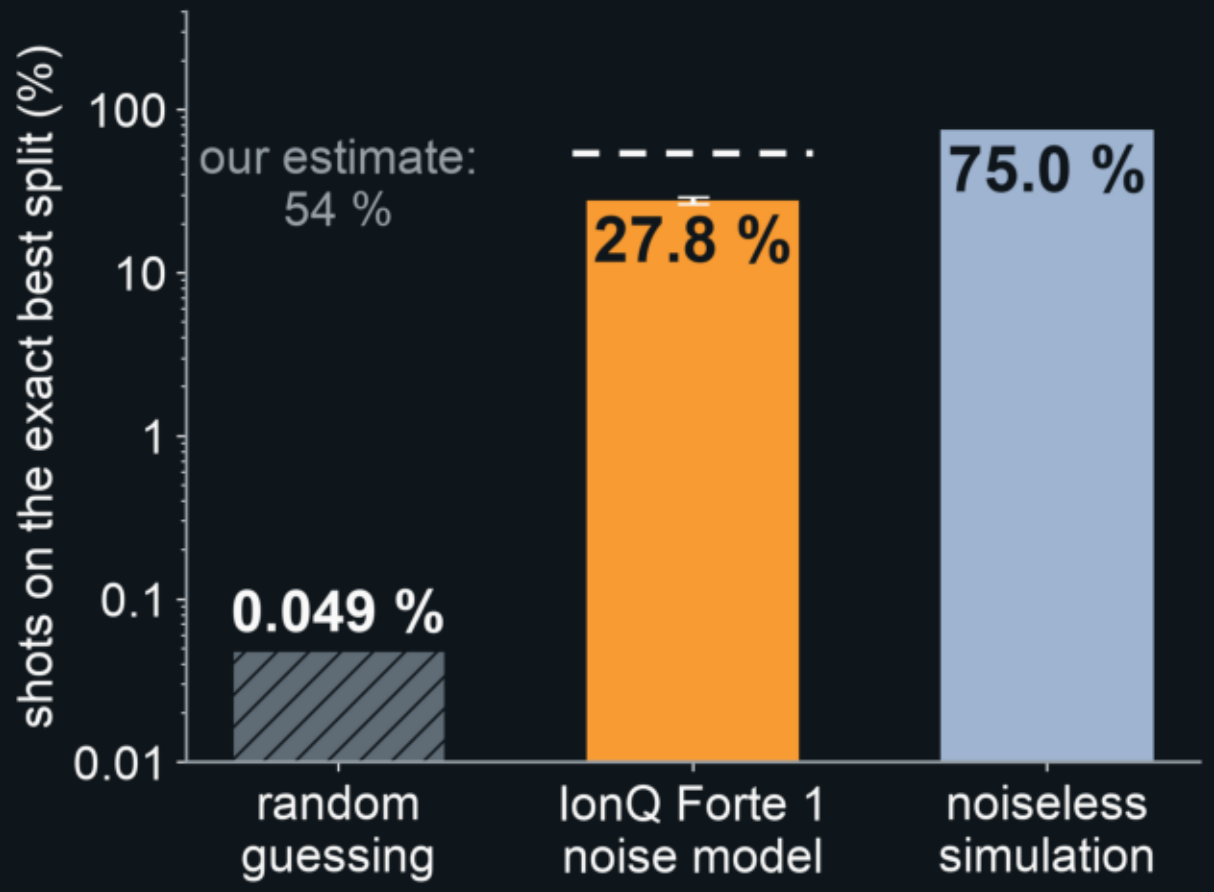
of 1,000 shots, about 570x random guessing

Most frequent answer: the exact optimum

IonQ Forte 1 noise model (Qollab), 11 Oct 2026:
varQITE ZY, 12 qubits, 66 two-qubit gates

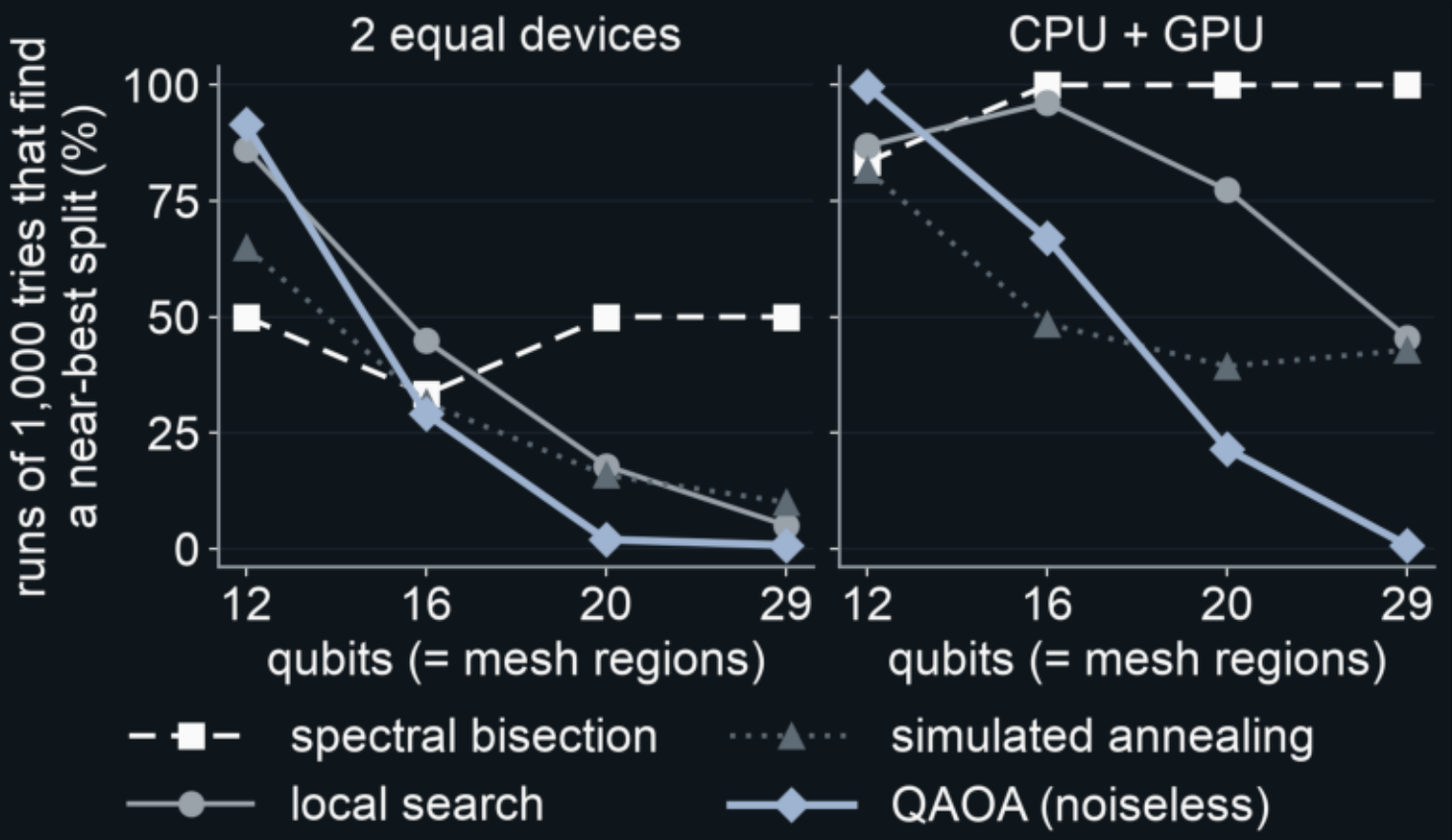
Noiseless: 75.0 %. Noise costs 47 points, about 2x what we estimated.

Classical, same 1,000 tries: annealing finds it in 50 % of runs, local search in 51 %.

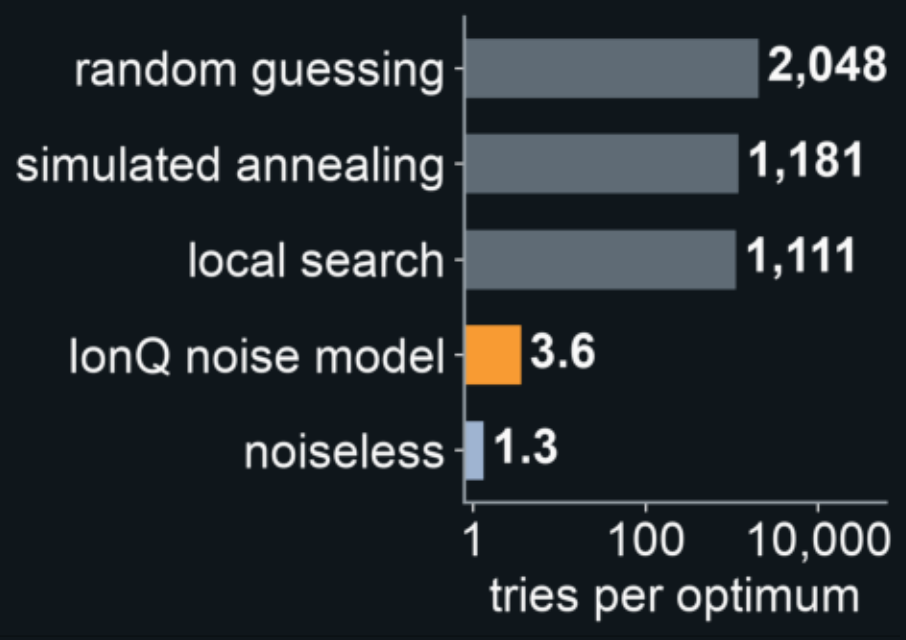


random and noiseless: noiseless simulation; classical: classical

Bigger meshes: classical methods win



server simulation, noiseless, 10 Oct 2026 (cases 2 and 9, seeds 7-9); QAOA = our best setting per size; the other lines are classical



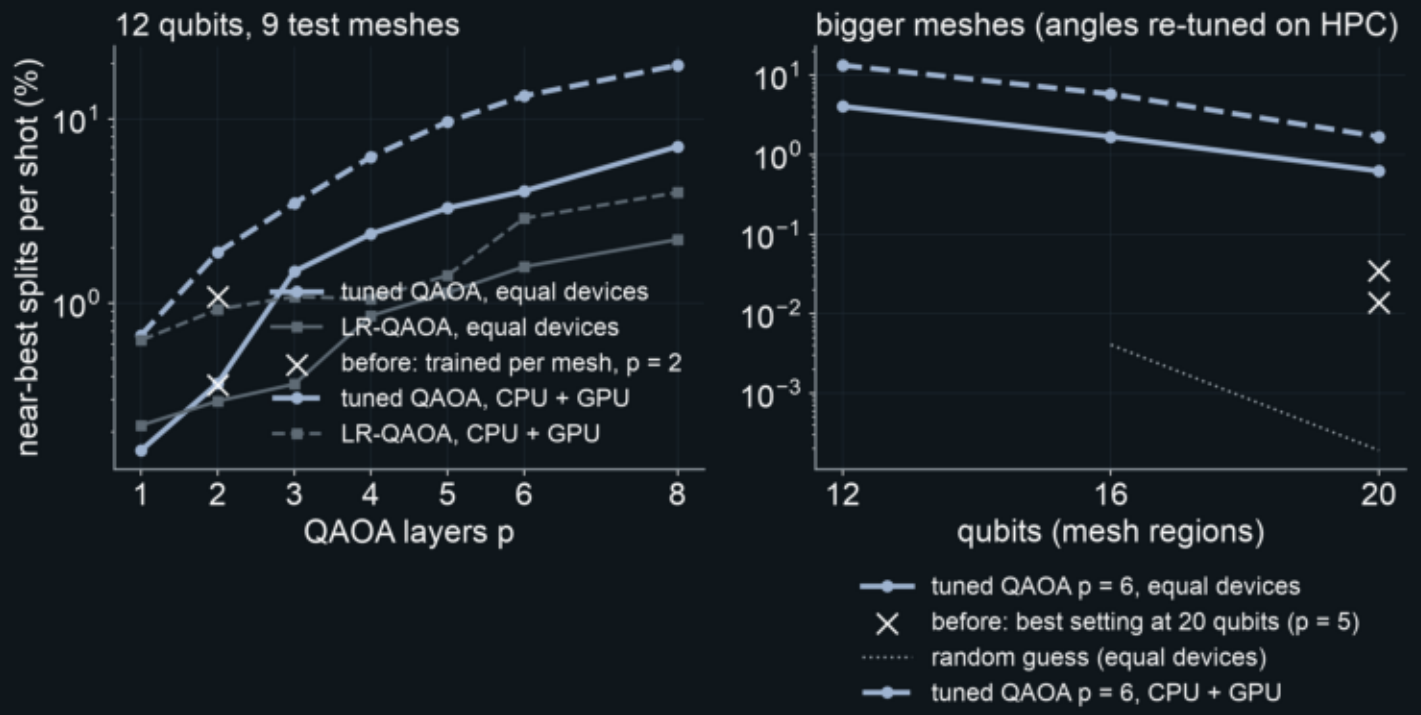
12 qubits: IonQ Forte 1 noise model (Qollab), classical

Caveats: varQITE first needed 7,581 noiseless training circuits. A shot is not the same work as a classical evaluation.

Tuning QAOA itself: about 20x more near-best answers

QAOA types we built

standard QAOA, all angles tuned
  (5 optimisers: COBYLA, Nelder-Mead, Powell, L-BFGS-B, SPSA)
LR-QAOA (linear ramp)
warm-start QAOA (spectral split)
CVaR-QAOA
symmetry-reduced QAOA
Iceberg-protected QAOA (error detection)
device-choice QAOA (many devices)



settings chosen on 18 separate calibration meshes, then frozen; scored on unseen meshes

12 qubits, 8 layers: 7.1 % / 19.5 % near-best (before 0.36 % / 1.08 %)
20 qubits, 6 layers: 0.62 % / 1.69 % (before 0.014 % / 0.035 %)

per shot, equal devices / CPU + GPU; noiseless simulation, HPC server, 11 Oct 2026; deeper circuits lose more under noise (0.5 % per gate estimate: best at 5 layers)

Run it on Qollab

```
Qollab > Run > IonQ Forte 1 (29q)
Running on: ionq_simulator
varqite_zy_equal x 1000 shots
P(optimum) measured 0.2780
x146  OPTIMUM  111100110000
x132  OPTIMUM  000011001111
[pending] live screenshot
```

Qollab project: [pending] Video: [pending]

GitHub: Oussema-t/zollerqu-qollab (branch mesh)

This effort is supported by Qollab & IonQ

Limits

1. A noisy simulator, not hardware.
2. CPU + GPU: the varQITE circuit fails. One qubit stays 50/50, and we proved why.
3. Beyond 12 qubits, classical methods win.

Next steps

1. Run the other circuits on the noise model.
2. Repair the circuit for unequal shares.
3. Split between more than two devices.

APPENDIX

Both device scenarios at 12 qubits

Scenario	Circuit (two-qubit gates)	Random	Noiseless	IonQ noise model
2 equal devices	varQITE ZY (66)	0.049 %	75.0 %	**27.8 %**
CPU + GPU	CVaR-QAOA, p = 2 (132)	0.024 %	0.58 %	[pending]
CPU + GPU	varQITE ZY (66)	0.024 %	0.00 % (cap proved)	[pending]

Share of shots that are the exact best split. The noisy runs for CPU + GPU are still to be made.

random, noiseless: noiseless simulation; last column: IonQ Forte 1 noise model (Qollab), 11 Oct 2026